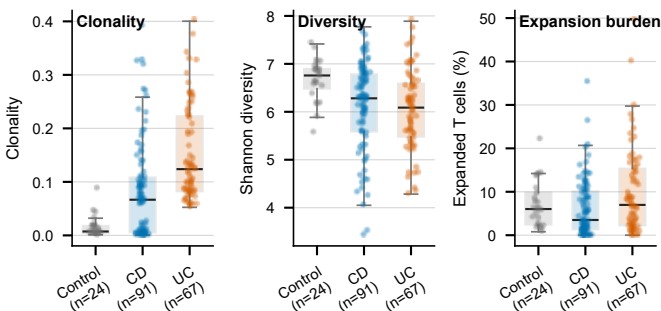
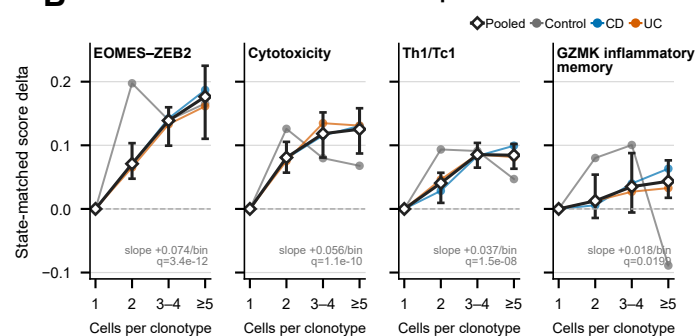


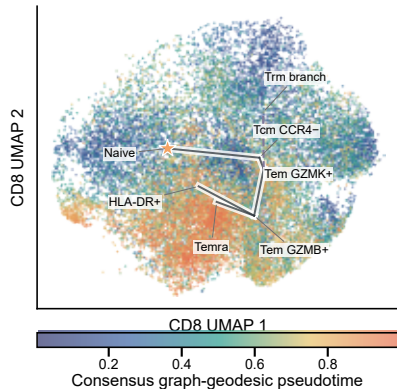
A Global TCR repertoire structure



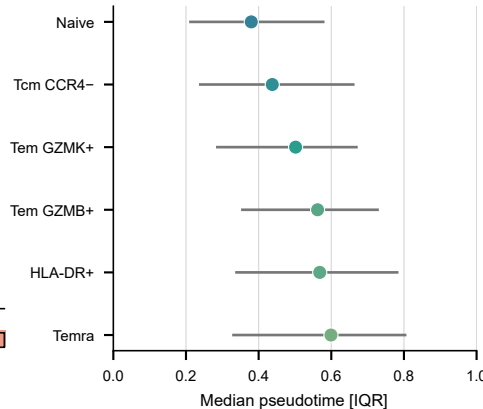
B State-dependent clone-size dose response



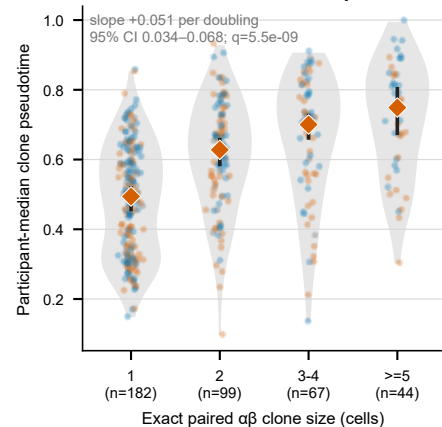
C Cross-sectional CD8 reference



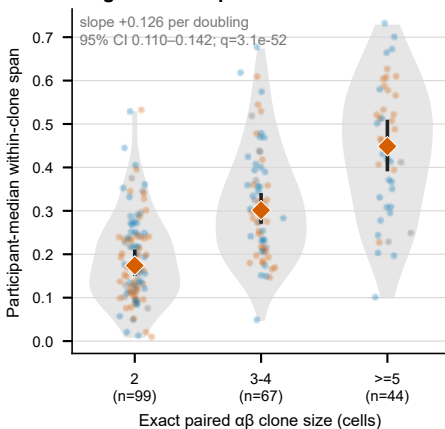
D Scalar-continuum positions



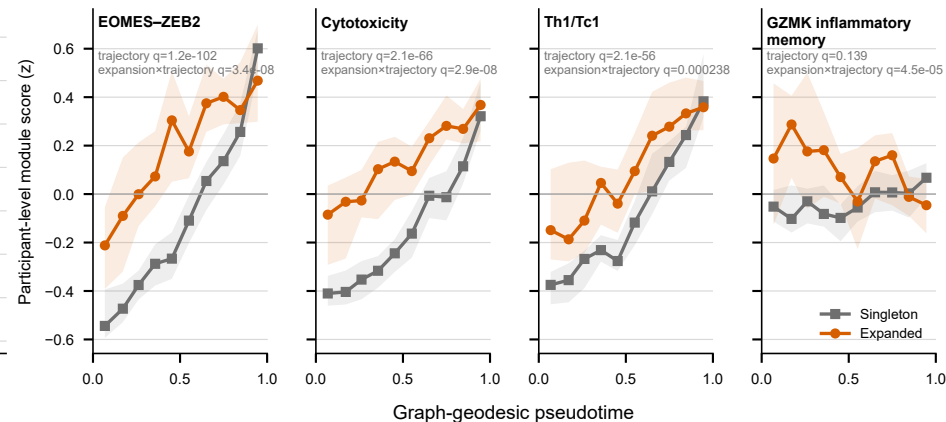
E Clone size tracks reference position



F Larger clones span the continuum



G Expansion-associated programs across pseudotime



H Independent longitudinal UC cohort, baseline

